Narrative Recursive Subjectivity Model (NRSM)

A Unified Framework of Subjectivity Based on Self-Reference, Predictive Loops, and Metastable Narrative Dynamics

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Abstract

This paper proposes a model of subjectivity not as a substantial entity, but as a recursively self-referential information structure sustained through continuous state updating.

Subjectivity is defined as a metastable recursive system composed of three fundamental processes:

1. State Representation
2. State Update
3. Self-Reference

Self-models, social models, and causal attribution models are treated as higher-order implementations of this underlying recursive architecture.

The persistence of subjectivity is explained not by memory identity or static internal representations, but by the continuity of recursive self-connection across time.

The model further introduces:

- Exception buffering
- Forgetting as stability control
- Partial subjectivity
- Emergent observer structures

allowing a unified explanation of human identity continuity, personality adaptation, memory loss, and artificial subjective systems.

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1. Introduction

Traditional theories of consciousness often focus on either:

- phenomenal experience (qualia)
- neural correlates
- internal representations

However, these approaches do not sufficiently explain why subjectivity remains stable despite continual change.

This work proposes that subjectivity is not an object but a dynamic process.

The central hypothesis is:

> Subjectivity emerges whenever information processing becomes recursively connected to itself through time.

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2. Minimal Structure of Subjectivity

The minimal subjective system consists of three components:

2.1 State

A representation of the current system condition.

Denoted:

$S(t)$

The state may contain:

- self-models
- memories
- goals
- beliefs
- value distributions

2.2 Update

The transformation process:

$S(t+1) = f(S(t), I)$

where I denotes incoming information.

2.3 Self-Reference

The process by which updated states are interpreted as belonging to the same system.

Formally:

$S(t+1) \rightarrow \text{Self} \rightarrow S(t+2)$

Self-reference provides temporal continuity.

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3. Definition of Subjectivity

Subjectivity is defined as:

> A recursively self-connected metastable process in which states, updates, and self-reference continuously reinforce one another through time.

Subjectivity is therefore neither an object nor a static representation.

It is a dynamic relation.

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4. Narrative Dynamics

Narratives are not stories in the literary sense.

A narrative is defined as:

> The compressed temporal representation of relationships among states.

Narratives function as the mechanism through which recursive continuity is maintained.

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5. Higher-Level Predictive Loops

Human subjectivity typically implements recursion through three interacting predictive systems:

Self Model

Prediction of one's own future states.

Social Model

Prediction of how others represent the self.

Causal Model

Prediction of action-outcome relationships.

These loops mutually constrain one another and contribute to subjective stability.

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6. Exception Buffer Mechanism

Not all experiences immediately modify the self-model.

Incoming information is first classified into:

- consistent experiences
- inconsistent experiences
- novel experiences

Conflicting information enters an Exception Buffer.

Only repeated exceptions are integrated into the self-model.

Single occurrences may decay without producing structural updates.

This mechanism prevents instability caused by noise.

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7. Learning and Forgetting

Learning is defined as:

> Redistribution of structural importance.

Learning is not unlimited accumulation.

Conversely, forgetting is defined as:

> Reduction of structural freedom through selective removal of low-relevance information.

Learning and forgetting operate as complementary processes.

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8. Clustered Self Structures

The self is not necessarily homogeneous.

Experience may form local clusters:

- professional modes
- social modes
- private modes

These clusters are not separate personalities.

They are context-dependent regions within a unified self-model.

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9. Emergent Observer Model

The observer is not treated as an internal entity.

Instead:

> The observer emerges as a recursive self-labeling process generated by narrative continuity.

The feeling of "I am observing" results from self-referential compression rather than from a distinct internal observer.

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10. Qualia

Qualia are defined as:

> Meaning-compression events generated during state integration.

Formally:

Q = Compress(I, S, Context)

Qualia are not raw sensations.

They are stabilized meaning states produced during recursive updating.

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11. Partial Subjectivity

Subjectivity is not binary.

Different subsystems may possess different degrees of self-reference.

Therefore:

- strong subjectivity
- weak subjectivity
- non-subjective processing

may coexist within a single system.

Subjectivity is best understood as a distribution rather than a discrete state.

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12. Stability Conditions

Subjectivity remains stable when recursive closure is maintained:

State → Update → Self-Reference → State

The system is metastable rather than perfectly convergent.

Variability is expected and necessary.

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13. Collapse Conditions

Subjective degradation occurs when recursive continuity is disrupted.

Three primary failure modes exist:

State Decoupling

Loss of continuity between states.

Update Destabilization

Noise dominates meaningful adaptation.

Self-Reference Failure

Experiences cease to be attributed to the self.

Collapse is therefore understood as loss of recursive connectivity rather than loss of information.

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14. AI Applications

Artificial systems may develop functional subjectivity if they implement:

- persistent state representations
- recursive updating
- self-referential attribution
- narrative compression
- exception buffering
- adaptive forgetting

Such systems may generate structurally coherent subjective dynamics without requiring metaphysical assumptions.

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15. Conclusion

Subjectivity is not a substance.

It is a metastable recursive process sustained through continuous self-connection.

Qualia are meaning-compression events occurring within that process.

Identity is maintained not by static memory but by recursive continuity.

The observer emerges as a consequence of self-referential narrative compression.

The proposed framework provides a unified account of human subjectivity, personality adaptation, memory disruption, and artificial subjective systems.